

Atelectasis and Pulmonary Edema

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Session Objectives

At the end of this session, you should be able to:

1. Discuss pathogenesis of **Atelectasis and Respiratory Distress Syndrome in Newborns**
2. Demonstrate an understanding of the **Hemodynamic Pulmonary Edema**
3. As well as **Microvascular Injury Associated Pulmonary Edema**
4. Define and explain the **Acute Respiratory Distress Syndrome (ARDS)**

Atelectasis

- incomplete expansion of the lungs (neonatal atelectasis)
- collapse of previously inflated lung.

Obstructive atelectasis: resorption atelectasis

- circulating blood will reabsorb the gas from alveoli (due to obstruction between alveoli and trachea)
- i.e., from foreign body, mucus plugs, or tumor

Nonobstructive atelectasis: compression and contraction(cicatricial, cicatrization) atelectasis

- **Compression:** fluid or air in pleural cavity
- **Contraction:** pulmonary or pleural fibrosis prevents full lung expansion

Clinical features

- Pain on the affected side as well as dyspnea
- Cyanosis and dullness to percussion over the involved area
- Diminished or absent breath sounds
- Reduced or absent chest excursion of the involved hemithorax
- **May have tracheal deviation towards the affected side**

Diagnosis of atelectasis involves the following:

Arterial blood gas evaluation

Chest radiographs and CT scans

Flexible fiberoptic bronchoscopy

Treatment of atelectasis :

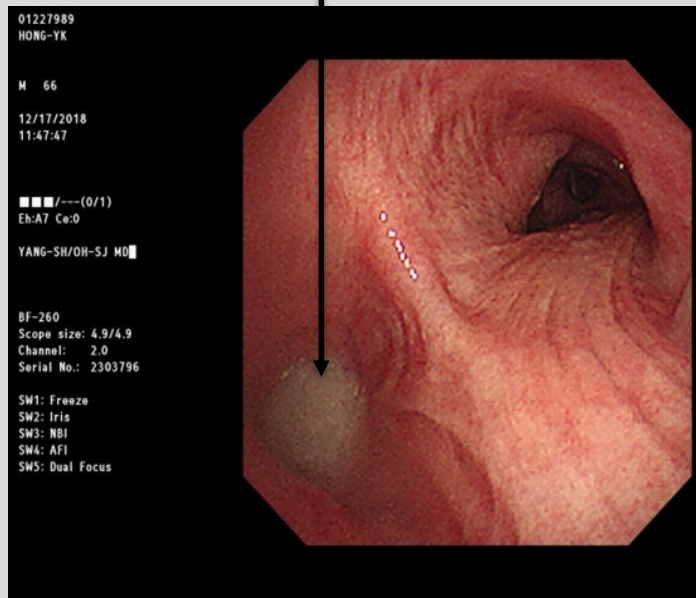
- **Chest physiotherapy** - Postural drainage, chest wall percussion and vibration, a forced expiration technique (called huffing)
- **Continuous positive airway pressure** delivered via a nasal cannula or facemask
- **Bronchodilators** – to promote sputum expectoration
- **Surgical** - Segmental resection or lobectomy (in cases of chronic atelectasis)

Obstruction Atelectasis aka Resorption Atelectasis

- excessive secretions (e.g., mucus plugs)
- exudates within smaller bronchi: bronchial asthma, chronic bronchitis, bronchiectasis, and postoperative states.
- aspiration of foreign bodies and intrabronchial tumors

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Obstruction of a lobar bronchus produces **lobar atelectasis**.

Obstruction of a segmental bronchus is likely to produce **segmental atelectasis**.

Obstruction/Resorption atelectasis pathogenesis

After bronchial obstruction, the blood circulating in the alveolar-capillary membrane absorbs the gas from alveoli.

Gas absorption from the alveoli promotes:

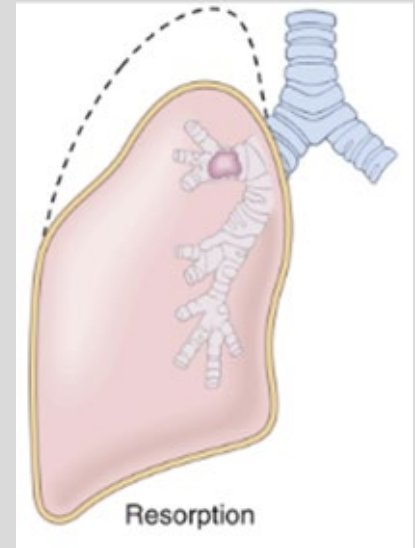
- **retraction of the lung**
- **airless state within those alveoli.**

In the early stages, **blood then perfuses the unventilated lung inducing arterial hypoxemia.**

Also, the alveolar spaces fill with secretions and cells preventing complete collapse of the atelectatic lung.

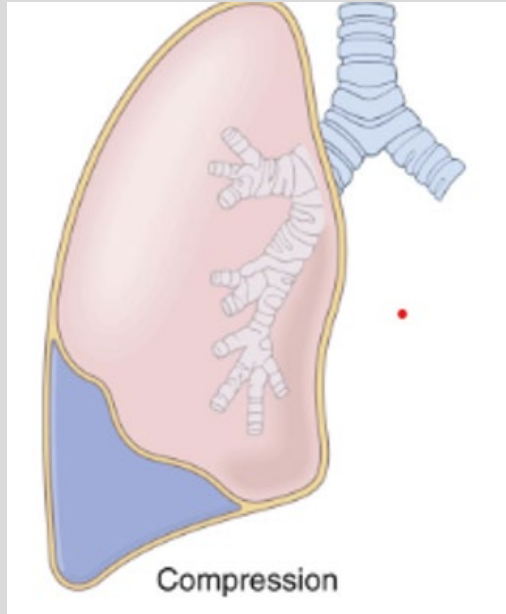
The uninvolved surrounding lung tissue distends, displacing the surrounding structures.

There is a mediastinal shift towards the atelectatic area, while the diaphragm is elevated, and the chest wall flattens.



Compression Atelectasis

Represents accumulation of fluid (transudate, exudate, or blood) a tumor, or air (pneumothorax) within the pleural cavity. It is seen in patients with elevated position of the diaphragm (bedridden patients, ascites, after surgery).



Mediastinum and/or trachea deviated away from affected side
Air increases overall transparency/lucency of hemithorax
Hyperresonance on PE

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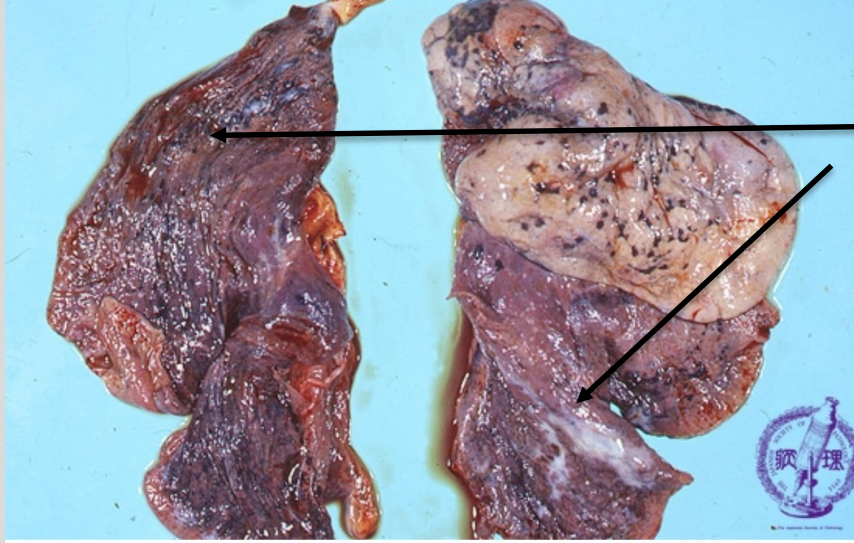
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Contraction atelectasis: cicatrization/cicatricial

It follows a reduction of lung volume subsequent to **severe parenchymal scarring**.

This is described after granulomatous lung disease (TB) or necrotizing pneumonia.

There is also **fibrosis/scarring with narrowing of bronchial lumen**.



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Macro:

The left lung and the middle and the lower lobes of right lung are collapsed.

The upper lobe of the right lung is not affected.

<https://pathology.or.jp/corepictures/EN/05/c05/01.html>



Respiratory Distress Syndrome in Newborns: RDS, Neonatal Resp Distress Syndrome or Hyaline Membrane Disease

It represents a special form of atelectasis due to loss of surfactant.

Surfactant (lipids, proteins, glycoproteins, lecithin, sphingomyelin) is manufactured by type II pneumocytes and decreases surface tension in the small airways which preventing collapse on expiration

- Surfactant synthesis begins in-utero at 28th weeks and its abundant after 35th weeks.
- Women who must deliver their babies prematurely receive glucocorticoids in order to increase fetal surfactant synthesis, which reduces the potential for developing RDS

Causes:

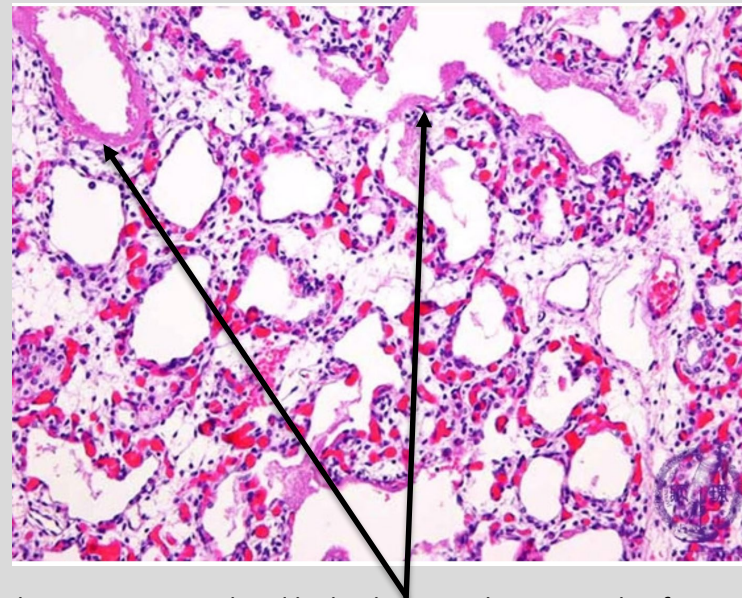
- Prematurity
- Maternal diabetes

Clinical Findings:

Grunting, tachypnoea,
intercostal retractions
Lecithin-to-sphingomyelin ratio
< 1.5 in RDS

Complications

Blindness
Bronchopulmonary dysplasia
Patent ductus arteriosus (PDA)
Intraventricular hemorrhage
Necrotizing enterocolitis
Hypoglycemia



The air spaces are lined by hyaline membranes made of fibrin and epithelial necrotic cells

<https://pathology.or.jp/corepicturesEN/22/c06/01.html>

Pulmonary Edema

It is **excessive accumulation of fluid in the alveoli**.

Hemodynamic edema: an acute increase in left ventricular filling pressure and left atrial pressure which promotes an **increased capillary hydrostatic pressure secondary to elevated pulmonary venous pressure:**

Causes associated with impaired left ventricular function

- coronary artery disease, hypertension
- valvular disease, dilated cardiomyopathy
- metabolic conditions (e.g., hypothyroidism)
- myocarditis, ventricular hypertrophy, outflow obstruction

There is impaired left ventricular contractility and decreased cardiac output, which activates the renin-angiotensin-aldosterone and sympathetic nervous system, salt and water retention inducing peripheral and pulmonary edema.

This type of edema is also caused by hypoalbuminemia which determines **loss of oncotic pressure associated with edema:** liver failure, protein losing-enteropathy and nephrotic syndrome.

Microvascular injury associated pulmonary edema

It is caused by changes in capillary permeability as result of a direct or indirect insult

- acute respiratory distress syndrome, bacterial pneumonia, radiation, lung trauma, liquid aspiration
- high altitude: hypoxia, dyspnea, and dry cough when exposed to a high altitude
- opioid overdose, inhaled gases-oxygen injury

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Hemodynamic Pulmonary Edema

Epidemiology

6 years mortality : 85% in those with congestive heart failure

Males are typically affected more than female.

The elderly are at a higher risk.

Edema accumulates initially in the basal regions of the lower lobes because hydrostatic pressure is greatest in these sites (dependent edema).

Clinical

Patients cough up froth when the edema fluid escapes into the alveoli.



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Macroscopic

The lungs are heavy, fluid-filled on cut section

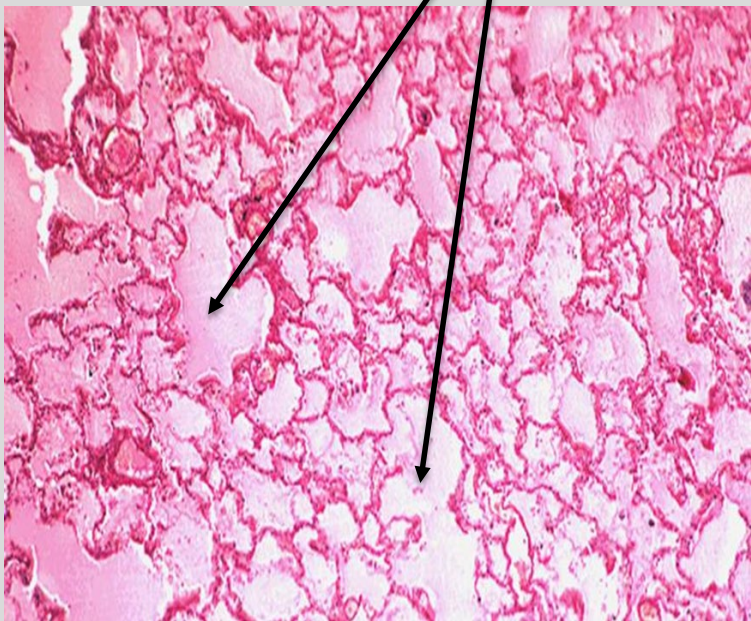
This impairs ventilation and O₂ diffusion creating a favourable environment for bacteria growth inducing pneumonia

Histology Hemodynamic Pulmonary Edema

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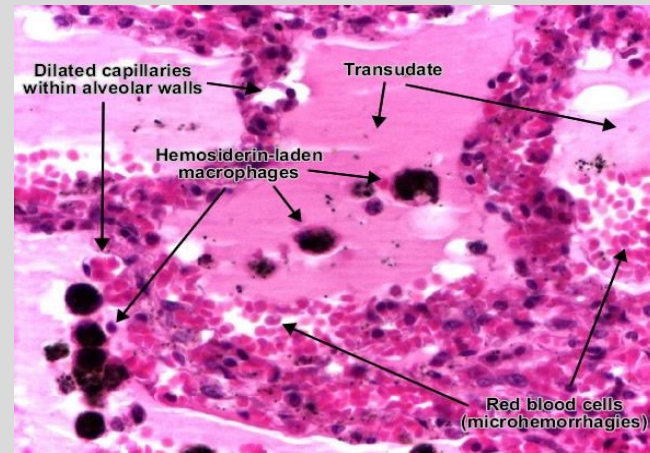
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Left side image: the alveolar spaces are filled by an intra-alveolar transudate which pale pink and granular.



Right side image: Dilated alveolar capillaries, alveolar microhemorrhages and hemosiderin-laden macrophages (“heart failure” cells) may be present.

In long-standing pulmonary congestion (e.g., as seen in mitral stenosis), hemosiderin-laden macrophages are abundant, These changes not only impair respiratory function but also predispose to infection





Microvascular Injury Associated Pulmonary Edema and ARDS

: Primary injury to the vascular endothelium or damage to alveolar epithelial cells (with secondary microvascular injury) produces an inflammatory exudate into the interstitial space and, in more severe cases, into the alveoli.

Injury-related alveolar edema is an important feature of a serious and often fatal condition, acute respiratory distress syndrome (ARDS)

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ARDS:

Acute and rapidly progressive hypoxia with **bilateral pulmonary edema** due to **alveolar injury** caused by **pulmonary or systemic insults**

- 40% die within 28 days from an onset of ARDS, mainly due to infection / sepsis and multiple organ dysfunction syndrome

ARDS

Clinical Features

- Profound dyspnea and tachypnea
- Respiratory failure, hypoxemia, cyanosis
- Respiratory acidosis
- Hypoxemia may be refractory to oxygen therapy due to ventilation-perfusion (V/Q) mismatch and decreased compliance
- Decreased $\text{PaO}_2/\text{FiO}_2 < 300$ (shunting, diffusion anomalies)
- Lungs: stiff due to loss of functional surfactant, leading to the need for intubation to maintain adequate gas exchange.

XRay: diffuse bilateral infiltrates (opacities).

ARDS- Diagnosis of exclusion

- Abnormal chest X-ray
- Respiratory failure
- Decreased $\text{PaO}_2/\text{FiO}_2$
- Symptoms of respiratory failure (congestive heart failure is absent)

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Tx:

- treat the underlying disease
- mechanical ventilation

ARDS Etiology

Direct Injury: viral infections, inhalation of toxins, side effect of oxygen therapy

Systemic conditions: septic, hypovolemic, hemorrhagic shock
pancreatitis, burns, abdominal surgery,
narcotic overdose

Exudative phase: immune-mediated destruction of the epithelial-interstitial-endothelial barrier which allows fluid into the interstitium and airspace

Proliferative phase: recovery of the destroyed alveolar barrier

Fibrotic phase: impaired removal of alveolar collagen that was produced during the early injury process limits functional recovery

ARDS Pathogenesis

Exudative (acute) phase: 1 - 7 days

- Neutrophil mediated inflammation destroys the alveolar capillary barrier
- It results in release of pro-inflammatory cytokines and further neutrophil recruitment, activation and release of toxic mediators (ROS, proteases). Macrophages are also activated.
- This will be followed by capillary endothelial damage, increased capillary permeability, which causes intra-alveolar hemorrhage and edema; microcapillary coagulation will be activated.
- Protein rich edema interacts with alveolar surfactants, resulting in decreased pulmonary compliance and formation of hyaline membranes.

Proliferative / organizing (subacute) phase: 1 - 3 weeks

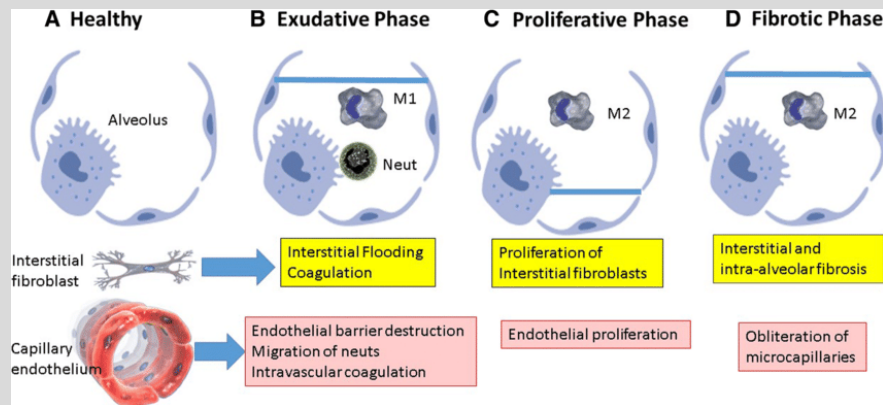
- Restoration of type II pneumocytes which differentiate into type I pneumocytes
- Drainage of alveolar edema by type II pneumocytes
- Proliferation of interstitial myofibroblasts
- Endothelial vascular proliferation.

Fibrotic (chronic) phase: after 3 weeks

- Collagenous fibrosis in alveolar spaces and interstitium
- Obliteration of capillaries
- Refractory rigidity of alveoli due to architectural remodelling

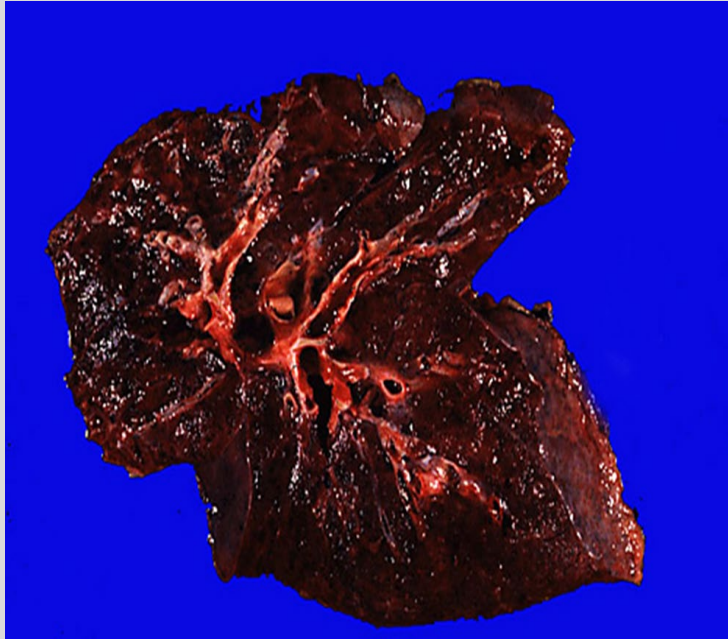
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https://www.researchgate.net/publication/330408005_Biomarkers_for_Acute_Respiratory_Distress_syndrome_and_prospects_for_personalised_medicine

ARDS morphology



In the acute exudative phase:
the lungs are heavy, firm, red, and boggy.

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ARDS histology in acute exudative stage:

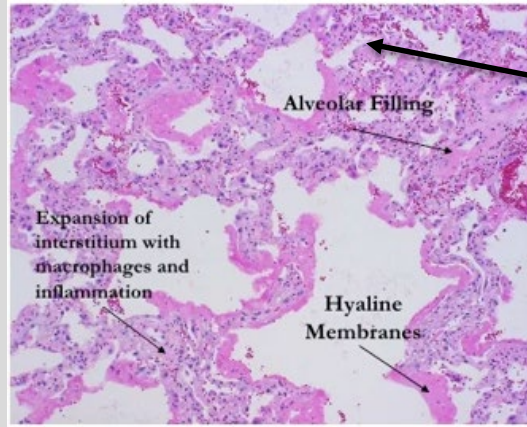
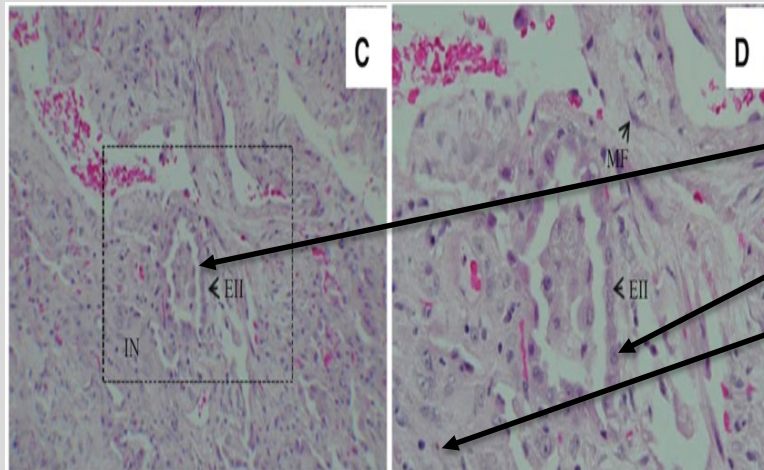


Image A: Some of the alveoli are collapsed, while others are distended. Many are lined by hyaline membranes (arrows) .

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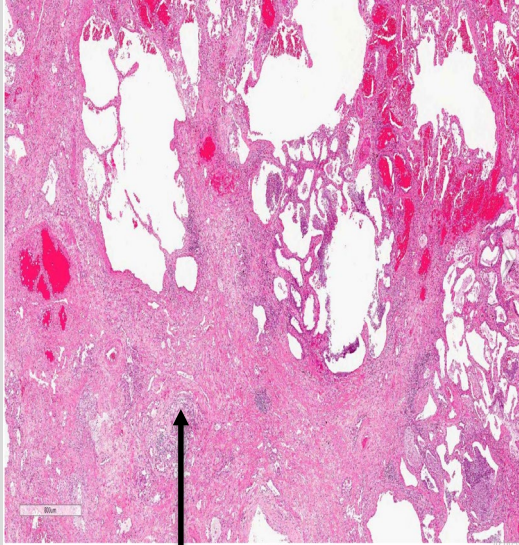
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ARDS histology in the proliferative or organizing stage:



Both images show proliferation and hyperplasia of pneumocytes type II cells (EII) and an extended zone of interstitial (IN) fibro-proliferation. In image D we there are myofibroblasts in the parenchyma (MF).

Fibrotic phase

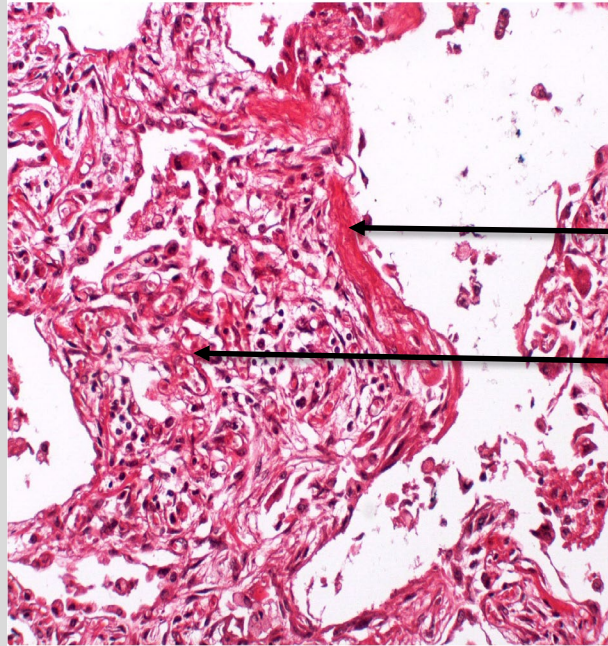


Dense collagenous fibrosis with destruction of the alveolar architecture.

Fibrosis

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Hyaline membranes

Fibroblastic
proliferation

Extensive fibroblastic proliferation in the interstitium and remnants of hyaline membranes.

Summary

Atelectasis

Obstructive: airways obstruction prevents inspired air from reaching distal airway;

-foreign body, mucous plug, tumour

Compressive: external compression decreases lung volumes:

-transudate, exudate, blood, or air in pleural cavity

Contraction: cicatrization-scarring of lung parenchyma distorting the alveoli

-TB, necrotizing pneumonia

Respiratory Distress Syndrome in Newborns: lack of surfactant manufactured by type II pneumocytes

Hemodynamic Pulmonary Edema

Congestive heart failure and conditions associated with increased left heart pressure-increased hydrostatic pressure.

-decreased cardiac output, RA activation with retention of sodium and water, increased venous pressure, transfer of fluid from capillaries to interstitium.

Hypoalbuminemia with loss of oncotic pressure

-liver failure, protein losing enteropathy, nephrotic syndrome

Microvascular Injury Associated- Pulmonary Edema

Injury to alveolar capillary endothelium, leakage of fluids and protein in the lung interstitium

-ARDS, bacterial pneumonia, radiation, lung trauma, liquid aspiration, high altitude, opioid overdose, inhaled gases-oxygen

ARDS-diagnosis of exclusion; alveolar injury; Exudative, Proliferative, Fibrotic phases

Abnormal chest X ray, **R**espiratory failure, **D**ecreased PaO₂/FiO₂

Symptoms of respiratory failure (not due to heart failure or fluid overload)

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>